(5) $y'(t) + 12y(t) = r(t)$, $r(t) = 2e^{10t}$, $y(0) = 0$

* initial conditions are zero

a) $y'(t) \xrightarrow{\mathcal{L}} sY(s) - y(0)$

$12y(t) \xrightarrow{\mathcal{L}} 12Y(s)$ $\quad r(t) \xrightarrow{\mathcal{L}} \frac{2}{s-10}$

$sY(s) - y(0) + 12Y(s) = \frac{2}{s-10} \rightarrow Y(s)(s+12) = \frac{2}{s-10}$ $\quad y(0) = 0$

$Y(s) = \left(\frac{2}{s-10}\right)\left(\frac{1}{s+12}\right) = \frac{2}{(s-10)(s+12)} = \frac{K_1}{s-10} + \frac{K_2}{s+12}$

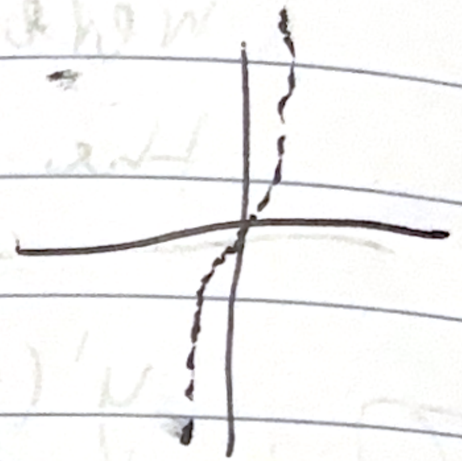
$K_1 = (s-10)\left[\frac{2}{(s-10)(s+12)}\right]_{s=10} = \frac{2}{10+12} = 1/11$

$K_2 = (s+12)\left[\frac{2}{(s-10)(s+12)}\right]_{s=-12} = \frac{2}{-12-10} = -1/11$

$Y(s) = \frac{1}{11(s-10)} + \frac{-1}{11(s+12)} \xrightarrow{\mathcal{L}^{-1}} \boxed{y(t) = \frac{e^{10t}}{11} - \frac{e^{-12t}}{11}}$

(6) b) $\lim_{s\to 0} sy(s) = \left[\frac{s}{11(s+10)} + \frac{-s}{11(s+12)}\right]_{s=0} = 0+0 = \boxed{0}$

c) $\lim_{t\to\infty} y(t) = \frac{e^{\infty} - e^{-\infty}}{11} = \frac{\infty}{11} = \infty$



Final value Theorem does not produce the same final value as $y(t)$